

# IMAT MOCK EXAM 7 — "The Real Thing"

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 38/90.0 · Mock exam

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## Section A - Logical Reasoning & Reading (Q1-9)

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**1.** "Resistance to antibiotics commonly arises when random mutations reduce a bacterium's sensitivity to a drug. Under antibiotic treatment, resistant cells survive while their susceptible competitors are killed, so resistant strains spread rapidly. Resistance genes can also move between bacterial species on plasmids. Long-term studies of sites where antibiotic use was halted years ago show that resistant strains sometimes remain common for a very long time, apparently because the fitness cost of resistance was offset by compensatory mutations."

Which statement is most strongly supported by the passage?

- A)** Antibiotics deliberately induce resistance mutations in bacteria.
- B)** Stopping antibiotic use does not guarantee that resistance will disappear.
- C)** Resistant bacteria are always fitter than susceptible ones.
- D)** Resistance genes cannot transfer between species.
- E)** Plasmids are a kind of virus.

**2.** A randomised trial reports that patients given drug X recovered in a median of 5 days versus 7 days with placebo, recovery being recorded by patient self-report. The authors conclude X speeds recovery.

Which of the following, if true, most weakens the authors' conclusion?

- A)** A few patients in each arm dropped out before day 7.
- B)** The placebo tablets were white and the X tablets were also white.
- C)** The trial lasted six months in total.
- D)** The trial ran in two countries at once.
- E)** Drug X is intensely bitter and clearly distinguishable from the tasteless placebo, so patients knew which arm they were in.

**3.** (General knowledge) The 2025 G7 summit was hosted by:

- A)** Japan
- B)** Italy
- C)** France
- D)** Canada
- E)** the United Kingdom

**4.** A clinic must schedule three locums, X, Y and Z, one for each of Monday, Tuesday and Wednesday. X must work earlier in the week than Z. Y cannot work on Tuesday.

Which of the following must be true?

- A)** X does not work on Wednesday.
- B)** X works on Monday.
- C)** Z works on Wednesday.
- D)** Y works on Monday.
- E)** Y works on Wednesday.

**5.** A chemical test turns blue if and only if protein X is present in a sample. A particular sample did not turn the test blue.

Which conclusion is valid?

- A)** The sample contains protein X.
- B)** The sample contains some other protein in quantity.
- C)** The sample does not contain protein X.
- D)** The test must have been performed incorrectly.
- E)** Protein X turns the test red.

6. A clock reads 9:18. What is the angle between the hour hand and the minute hand?  
**A)**  $144^\circ$  **B)**  $162^\circ$  **C)**  $171^\circ$  **D)**  $174^\circ$  **E)**  $180^\circ$
7. Of 1,000 people screened, 100 have the disease. The test is positive in 90 of the 100 diseased people, and also positive in 180 of the 900 healthy people.  
 What fraction of all positive results are false positives?  
**A)**  $1/3$  **B)**  $1/5$  **C)**  $1/2$  **D)**  $2/3$  **E)**  $9/10$
8. A cyclist rides from A to B at 60 km/h and returns immediately along the same route at 120 km/h. The average speed for the whole trip is:  
**A)** 60 km/h **B)** 75 km/h **C)** 80 km/h **D)** 90 km/h **E)** 100 km/h
9. A sequence begins 2, 3, 5, 9, 17, 33, ... Each term follows one rule from the previous term. What is the next term?  
**A)** 49 **B)** 63 **C)** 64 **D)** 65 **E)** 66

## Section B - Biology (Q10-32)

10. A plant with genotype AaBb for two unlinked genes is crossed with a double-recessive aabb plant, and the offspring phenotypes appear 1:1:1:1. This result shows that the AaBb parent:  
**A)** produces only two kinds of gametes  
**B)** must have undergone a mutation during meiosis  
**C)** has its two genes tightly linked on one chromosome  
**D)** is homozygous dominant at both genes  
**E)** is heterozygous at both genes and passes the four allele combinations into gametes equally
11. In snapdragons, flower colour shows incomplete dominance: CRCR is red, CRCW is pink, CWCW is white. Two pink plants are crossed. The expected offspring ratio is:  
**A)** 1 red : 2 pink : 1 white  
**B)** all pink  
**C)** 9:3:3:1  
**D)** 3 red : 1 white  
**E)** 1 red : 1 white
12. Two genes assort independently. A cross AaBb  $\times$  aabb is performed. What fraction of the offspring is expected to be aabb?  
**A)** 0 **B)**  $1/16$  **C)**  $1/4$  **D)**  $1/8$  **E)**  $1/2$
13. Two unaffected parents have a child affected by a rare autosomal recessive disease. What is the probability that their next child is affected?  
**A)** 0 (immunity has developed)  
**B)**  $1/2$   
**C)**  $1/4$   
**D)**  $1/8$   
**E)**  $2/3$
14. On the lagging strand, DNA polymerase works away from the replication fork, producing short Okazaki fragments. Which enzyme joins these fragments into a continuous strand?  
**A)** helicase  
**B)** DNA ligase  
**C)** single-strand binding protein  
**D)** primase  
**E)** topoisomerase
15. A DNA template strand contains the triplet 3'-TAC-5'. The mRNA produced from this triplet is:  
**A)** 5'-AUG-3' **B)** 5'-ATG-3' **C)** 5'-GUA-3' **D)** 5'-CAT-3' **E)** 5'-UAC-3'

- 16.** Which of the following codons signals termination of translation?  
**A)** UGG **B)** UGA **C)** GGG **D)** AUG **E)** UAC
- 17.** A protein is heated to 70 °C and denatures. What has happened to its structure?  
**A)** The protein has become a different protein.  
**B)** Individual amino acids have been chemically destroyed.  
**C)** The primary structure has been changed.  
**D)** Peptide bonds along the backbone have broken.  
**E)** Weak interactions (hydrogen bonds, ionic and hydrophobic contacts) have been disrupted while the amino-acid sequence is preserved.
- 18.** An inhibitor reduces an enzyme's  $V_{max}$  by 75% but leaves  $K_m$  exactly unchanged. The inhibitor is best described as:  
**A)** competitive  
**B)** a substrate analogue competing for the active site  
**C)** a cofactor  
**D)** an allosteric activator  
**E)** non-competitive
- 19.** A poison blocks complex I of the mitochondrial electron transport chain. Which statement follows?  
**A)** Pyruvate oxidation accelerates to compensate.  
**B)** Oxidation of NADH halts, but electrons from  $FADH_2$  can still enter at complex II and reach oxygen.  
**C)** All electron transport through complexes II, III and IV stops immediately and completely.  
**D)** Glycolysis stops.  
**E)** ATP synthase reverses and consumes water.
- 20.** The Calvin cycle cannot keep running for long in darkness because it directly requires:  
**A)** ATP and NADPH supplied by the light reactions  
**B)** glucose from starch breakdown  
**C)** chlorophyll excitation  
**D)** oxygen released by photosystem II  
**E)** carbonic acid
- 21.** A red blood cell is placed in seawater, which is hypertonic to blood plasma. What happens?  
**A)** Nothing: cells are always in osmotic balance.  
**B)** The cell divides.  
**C)** Sodium ions are pumped out to compensate, so volume is constant.  
**D)** Water leaves the cell and it shrinks (crenates).  
**E)** Water enters the cell and it swells.
- 22.** The protein p53 accumulates when DNA is damaged in G1. Its role is to:  
**A)** replicate the damaged DNA segment  
**B)** halt the cell cycle so repair can occur, triggering apoptosis if damage is unrepairable  
**C)** move the cell into S phase regardless of damage  
**D)** phosphorylate the nuclear envelope  
**E)** push the cell directly into mitosis
- 23.** Nondisjunction occurs during meiosis in a human oocyte, producing an egg with 24 chromosomes. Fertilisation by a normal sperm gives a zygote with:  
**A)** 47 chromosomes (a trisomy, such as trisomy 21 if chromosome 21 is involved)  
**B)** certain death of the zygote in all cases  
**C)** 46 chromosomes, normal  
**D)** 45 chromosomes (monosomy)  
**E)** 46 chromosomes with a deletion

- 24.** The rapid rising (depolarisation) phase of a nerve action potential is caused by:
- A)** the sodium-potassium pump running forward at maximum speed
  - B)** chloride ions leaving the axon
  - C)** the membrane becoming impermeable to all ions
  - D)** opening of voltage-gated potassium channels,  $K^+$  entering
  - E)** opening of voltage-gated sodium channels,  $Na^+$  rushing in
- 25.** At a chemical synapse, the immediate trigger for neurotransmitter release from the presynaptic terminal is:
- A)** reuptake of neurotransmitter
  - B)** arrival of the neurotransmitter enzyme
  - C)** influx of sodium into the postsynaptic cell
  - D)** influx of calcium ions into the terminal
  - E)** exit of potassium from the terminal
- 26.** Several hours after a carbohydrate-rich meal, blood glucose is falling. Which endocrine change is occurring?
- A)** insulin rises, glucagon rises
  - B)** insulin falls, glucagon falls
  - C)** insulin rises, glucagon falls
  - D)** insulin falls, glucagon rises
  - E)** neither hormone changes
- 27.** Antivenom is produced by immunising horses, then collecting and purifying the antibodies. A snakebite victim receiving antivenom acquires:
- A)** active natural immunity
  - B)** no immunity, since antibodies cannot be injected
  - C)** immunity produced by the victim's own plasma cells weeks later
  - D)** active artificial immunity with lasting memory
  - E)** passive artificial immunity: ready-made antibodies, protection that is immediate but without immune memory
- 28.** A patient is blood group O, Rh-negative. Which donor red-cell type can be transfused to this patient without a risk of agglutination (within the ABO and Rh systems)?
- A)** A negative
  - B)** AB negative
  - C)** O negative
  - D)** O positive
  - E)** any type, as long as it is Rh-negative
- 29.** Pancreatic lipase, acting in the small intestine, digests:
- A)** vitamins into coenzymes
  - B)** triglycerides (fats) into fatty acids and glycerol
  - C)** starch into maltose
  - D)** nucleic acids into nucleotides
  - E)** proteins into amino acids
- 30.** Glucose that is filtered at the glomerulus is normally:
- A)** almost completely reabsorbed in the proximal convoluted tubule
  - B)** excreted entirely in the urine
  - C)** converted to urea in the nephron
  - D)** secreted into the filtrate by the distal tubule
  - E)** reabsorbed only in the collecting duct under ADH control

31. Which cells deposit new bone matrix during bone growth and remodeling?  
A) osteoclasts  
B) chondrocytes  
C) red marrow stem cells  
D) osteocytes in lacunae  
E) osteoblasts
32. Cells are briefly exposed to radioactively labelled thymidine. Radioactivity will be incorporated into DNA during which phase?  
A) G0 B) G1 C) M D) S E) G2

### Section C - Chemistry (Q33-47)

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33. What are the oxidation numbers of nitrogen in the ammonium ion ( $\text{NH}_4^+$ ) and in the nitrate ion ( $\text{NO}_3^-$ ), respectively?  
A) -5 and +3  
B) -3 and +5  
C) -3 and -3  
D) +1 and +1 (the average of the two nitrogens)  
E) +5 and +5
34. Given the activity series  $\text{Mg} > \text{Zn} > \text{Cu} > \text{Ag}$ , which reaction occurs spontaneously at room temperature?  
A)  $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$   
B) none of these reactions occurs spontaneously  
C)  $\text{Cu(s)} + \text{Mg}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Mg(s)}$   
D)  $\text{Cu(s)} + \text{Zn}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Zn(s)}$   
E)  $\text{Ag(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Ag}^+(\text{aq}) + \text{Cu(s)}$
35. Which bond is the most polar (largest electronegativity difference)? (EN: H 2.2, C 2.55, N 3.04, O 3.44)  
A) C-N B) C-O C) O-H D) C-H E) N-H
36. Which molecule contains polar bonds but has zero net dipole moment?  
A)  $\text{CHCl}_3$  B)  $\text{CO}_2$  C)  $\text{H}_2\text{O}$  D)  $\text{SO}_2$  (bent) E)  $\text{NH}_3$
37. 100 mL of 0.1 M HCl is mixed with 100 mL of 0.3 M NaOH. The pH of the resulting solution is:  
A) 1 B) 2 C) 7 D) 12 E) 13
38. An aqueous solution of HCl is prepared at the extraordinarily low concentration of  $1.0 \times 10^{-8}$  M. At 25 °C its pH is:  
A) exactly 7  
B) undefined, because the concentration is below  $10^{-7}$   
C) slightly below 7  
D) slightly above 7  
E) about 2
39. 25.0 mL of  $\text{H}_2\text{SO}_4$  solution is completely neutralised by 20.0 mL of 0.15 M NaOH ( $\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$ ). The molarity of the acid is:  
A) 0.03 M B) 0.06 M C) 0.12 M D) 0.15 M E) 0.30 M
40. 2.50 g of a hydrate of copper(II) sulfate ( $\text{CuSO}_4 \cdot n\text{H}_2\text{O}$ ,  $M(\text{CuSO}_4) = 160$  g/mol) is heated to constant mass, leaving 1.60 g of anhydrous  $\text{CuSO}_4$ . The value of n is:  
A) 1 B) 2 C) 3 D) 4 E) 5
41. What is the mass of 5.6 L of oxygen gas  $\text{O}_2$  measured at STP (molar volume 22.4 L/mol;  $M(\text{O}_2) = 32$  g/mol)?  
A) 2 g B) 4 g C) 8 g D) 16 g E) 32 g

42. The species  $\text{Na}^+$ ,  $\text{Mg}^{2+}$ ,  $\text{F}^-$ ,  $\text{O}^{2-}$  and Ne all have 10 electrons. Which is the smallest?  
**A)**  $\text{Mg}^{2+}$  **B)**  $\text{F}^-$  **C)**  $\text{Na}^+$  **D)** Ne **E)**  $\text{O}^{2-}$
43. Which of the following can exist as cis and trans isomers?  
**A)** ethene  
**B)** but-1-ene  
**C)** propane  
**D)** but-2-ene  
**E)** 2-methylpropene
44. Propan-1-ol is oxidised with acidified potassium dichromate, and the volatile organic product is distilled off as it forms. The product is:  
**A)** propane **B)** propanone **C)** propan-2-ol **D)** propanal **E)** propanoic acid
45. For the equilibrium  $\text{A} \rightleftharpoons \text{B}$ , a solution initially containing only A at 1.0 M reaches equilibrium with  $[\text{B}] = 0.75 \text{ M}$ . The equilibrium constant  $K_c$  is:  
**A)** 0.25 **B)** 0.75 **C)** 1.33 **D)** 3.0 **E)** 4.0
46. For an exothermic reaction at equilibrium, increasing the temperature:  
**A)** decreases  $K_c$   
**B)** increases  $K_c$   
**C)** leaves  $K_c$  unchanged  
**D)** increases  $K_c$ , which then returns to its old value  
**E)** has an unpredictable effect without the numerical  $K_c$
47. What is the approximate molarity of pure water at 25 °C (density 1.0 g/mL,  $M(\text{H}_2\text{O}) = 18 \text{ g/mol}$ )?  
**A)** 0.056 M **B)** 1.0 M **C)** 18 M **D)** 55.6 M **E)** 100 M

### Section D - Mathematics (Q48-54)

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48. Solve  $x^2 - 5x + 6 > 0$ .  
**A)** all real  $x$  **B)**  $-3 < x < -2$  **C)**  $x < 2$  or  $x > 3$  **D)**  $x < 2$  only **E)**  $2 < x < 3$
49. Solve  $|2x - 1| \leq 5$ .  
**A)** no solution **B)**  $-2 \leq x \leq 3$  **C)**  $x \geq -2$  **D)**  $-3 \leq x \leq 2$  **E)**  $x \leq 3$
50. Given  $\log_{10} 2 = 0.30$ , the value of  $\log_{10} 5$  is:  
**A)** 0.50 **B)** 0.60 **C)** 0.70 **D)** 1.30 **E)** 2.30
51. Two fair dice are rolled. What is the probability of getting at least one six?  
**A)** 1/6 **B)** 1/12 **C)** 1/36 **D)** 5/36 **E)** 11/36
52. A vertical pole 6 m tall casts a shadow when the sun's elevation is 45°. How long is the shadow? ( $\tan 45^\circ = 1$ ,  $\tan 30^\circ = 0.58$ )  
**A)** 3 m **B)** 4.2 m **C)** 6 m **D)** 8.5 m **E)** 12 m
53. If  $f(x) = (2x + 3)/5$ , then  $f^{-1}(x)$ , the inverse function, is:  
**A)**  $(5x - 3)/2$  **B)**  $2x + 3/5$  **C)**  $(2x - 3)/5$  **D)**  $(5x + 3)/2$  **E)**  $5/(2x + 3)$
54. What is the vertex (turning point) of the parabola  $y = x^2 - 4x + 7$ ?  
**A)** (2, 3) **B)** (-2, 3) **C)** (2, 11) **D)** (4, 7) **E)** (-2, 19)

### Section E - Physics (Q55-60)

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55. A 12 V battery drives a 4  $\Omega$  resistor in series with a parallel pair of two 4  $\Omega$  resistors. What power is dissipated in one of the two parallel resistors?  
**A)** 2 W **B)** 4 W **C)** 8 W **D)** 16 W **E)** 24 W

- 56.** A stone is dropped from rest from a height of 5 m ( $g = 10 \text{ m/s}^2$ , air resistance negligible). Its speed just before impact is:  
**A)** 5 m/s   **B)** 10 m/s   **C)** 20 m/s   **D)** 25 m/s   **E)** 50 m/s
- 57.** 1 kg of water and 1 kg of iron each receive 10 kJ of heat. ( $c_{\text{water}} = 4200$ ,  $c_{\text{iron}} = 450 \text{ J/(kg}\cdot\text{K)}$ .) Which statement is correct?  
**A)** It depends on the shape of the containers.  
**B)** Neither changes temperature.  
**C)** The water ends at the higher temperature.  
**D)** Both end at the same temperature.  
**E)** The iron ends at the higher temperature, rising about 22 K versus about 2.4 K for the water.
- 58.** A ball falls freely from rest for 2.0 s ( $g = 10 \text{ m/s}^2$ ). The distance it falls is:  
**A)** 5 m   **B)** 10 m   **C)** 20 m   **D)** 40 m   **E)** 80 m
- 59.** A person stands on bathroom scales inside a lift that is accelerating upward. Compared with the person's true weight:  
**A)** the reading depends on the lift's velocity, not its acceleration  
**B)** the scale reads zero  
**C)** the scale reads less  
**D)** the scale reads exactly the true weight  
**E)** the scale reads more, because the floor must supply  $m(g + a)$
- 60.** A sound wave of frequency 500 Hz has a wavelength of 0.68 m in air. The speed of sound is:  
**A)** 34 m/s   **B)** 170 m/s   **C)** 340 m/s   **D)** 500 m/s   **E)** 680 m/s

## Answer key

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This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1  | B | 2  | E | 3  | D | 4  | A | 5  | C |
| 6  | C | 7  | D | 8  | C | 9  | D | 10 | E |
| 11 | A | 12 | C | 13 | C | 14 | B | 15 | A |
| 16 | B | 17 | E | 18 | E | 19 | B | 20 | A |
| 21 | D | 22 | B | 23 | A | 24 | E | 25 | D |
| 26 | D | 27 | E | 28 | C | 29 | B | 30 | A |
| 31 | E | 32 | D | 33 | B | 34 | A | 35 | C |
| 36 | B | 37 | E | 38 | C | 39 | B | 40 | E |
| 41 | C | 42 | A | 43 | D | 44 | D | 45 | D |
| 46 | A | 47 | D | 48 | C | 49 | B | 50 | C |
| 51 | E | 52 | C | 53 | A | 54 | A | 55 | B |
| 56 | B | 57 | E | 58 | C | 59 | E | 60 | C |